

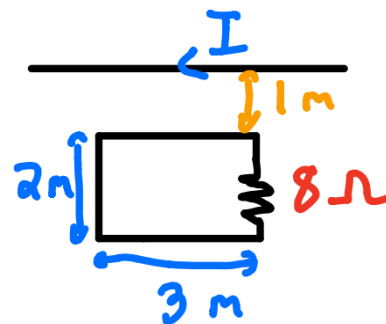


Directions: Show your fluxing work.

1. A rectangular loop of dimension 2 m x 3 m is in the plane of the page as shown. A long straight wire also in the plane of the page carries a current I in the direction shown, with I varying in time as given by $I(t) = 3 + 3t$. The wire is 1.0 m from the loop.

- Calculate the magnetic flux through the rectangular loop at time $t = 0$ s.
- What direction is the current induced in the loop?
- Determine the value of the current in the loop at time $t = 3$ s.

a) $\frac{9\mu_0}{2\pi} \ln|3|$ b) CW c) $\frac{9\mu_0}{16\pi} \ln|3|$



2. A conducting rod is free to move on a pair of horizontal, frictionless conducting rails a distance $L = 20$ cm apart. The rails are connected at one end so a complete circuit is formed. The rod has a mass $m = 0.2$ kg, the resistance of the circuit is $R = 20 \Omega$, and there is a uniform magnetic field of magnitude $|\vec{B}| = 2$ T, directed perpendicularly into the plane of the rails, as shown. At time $t = 0$, the rod has a speed $v = 20 \frac{m}{s}$ to the left.

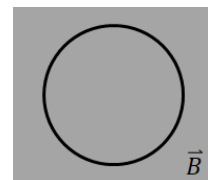


- Determine the direction and magnitude of the net magnetic force on the rod at time $t = 0$.
- Using integral calculus, determine a function for the speed v of the rod as a function of time t .
- Calculate the total energy dissipated by the resistor beginning at $t = 0$.

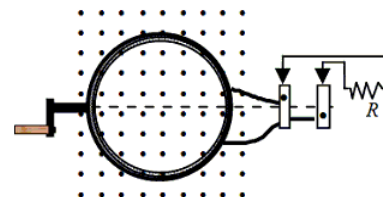
a) Force is to the right, $F = 0.16$ N
b) $v(t) = 20e^{-.04t}$
c) 40 J

3. A wire loop of area A is placed in spatially uniform magnetic field that is perpendicular to the plane of the loop. The induced emf in the loop is given by $\varepsilon(t) = kA\sqrt{t}$, where k is a constant. Write an expression for the magnetic field as a function of time.

$\frac{2}{3}kt^{\frac{3}{2}}$

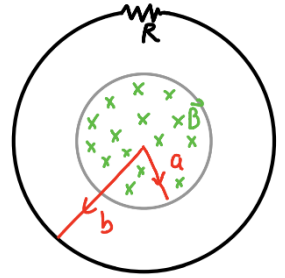


4. A circular coil has 305 turns and a radius of 0.05 m. The coil is used as an ac generator by rotating it in a 0.600 T magnetic field, as shown in the figure. At what angular speed should the coil be rotated so that the maximum emf is 135 V?



94 1/s

5. A magnetic field exists in a circular region of space with a radius of $a = 0.5 \text{ m}$, as shown. At time $t = 0$, the magnetic field is 0, but increasing at a rate of 0.2 T/s . A loop of wire of radius $b = 0.8 \text{ m}$ is concentric with the field and has a resistor with a resistance of $R = 7 \Omega$.



- Determine the magnitude and direction of the current in the resistor.
- Determine the rate at which heat is being produced in the resistor.

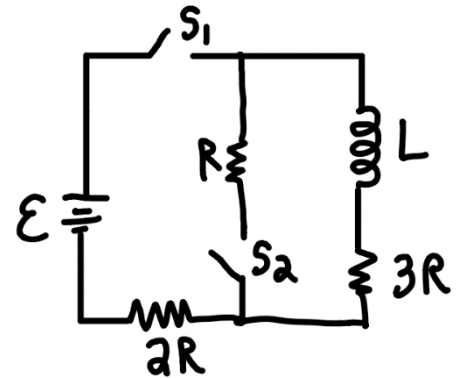
- .011 A, CCW
- .0008 W

6. A 80 loop solenoid has a radius of 0.1 m and a length of 0.3 m .

- Calculate the inductance of the solenoid. Show your work.
- The solenoid is connected to a battery with a variable emf given by $\varepsilon(t) = 2\sin(\pi t)$. Determine an expression for the current in the inductor.

- .0008 H
- 755 A/s

7. There's a circuit on the right. $\varepsilon = 100 \text{ V}$, $R = 50 \Omega$, $L = 8 \text{ H}$. S_1 is initially closed at time $t = 0 \text{ s}$ while S_2 is kept open.

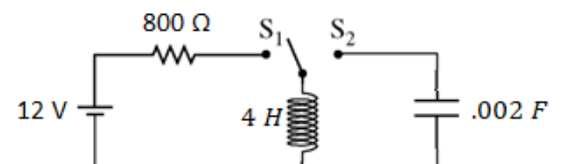


- Determine the rate of change of current in the inductor when the current in the battery is 0.1 A .
- Using integral calculus, determine an expression for the current in the inductor starting at $t = 0 \text{ s}$.
- After a long time, S_1 is opened while S_2 is closed. Determine the initial current (magnitude and direction) in resistor $3R$.

- 9.375
- $I(t) = 0.4(1 - e^{-\frac{5}{2}t})$
- 0.4 A down

8. An LC circuit is set up as shown. At time $t = 0$, the switch is set to position S_1 .

- Calculate the time for the voltage across the resistor to reach -6 V .
- After oh so long, the switch is set to position S_2 .



- Explain why the circuit will oscillate.
- Set up a differential equation to determine the charge q , in the capacitor once the switch is flipped. Use your equation to determine the period of oscillation of the LC circuit.
- Calculate the charge in the capacitor when current in the inductor is changing at a rate of 1 A/s .

- .003 s
- i. dude ii. no, 0.56 s iii. .008